Integration Consistency Checklist

Step 6 — Apex Insurance Group Integrated Architecture Package

Project	Apex Insurance Group Integrated Architecture and Sign-off Package
Template Section	Step 6 — Integration Consistency Verification (Sections 1 to 4)
Learner Name	Dennis Lam
Deliverable	Verification record accompanying Sections 1 to 4

Result: all three required checks pass. Grounding sources: 12 of 12 in the agent's own Project. Handoff fields: 19 of 19 defined in the Section 2 output schema. Failure nodes: 3 of 3 traced to Section 2 agents and Section 3 handoffs. Additional cross-section checks: 14 of 14 pass.

One inconsistency was found while running the checks and fixed before this record was produced (item 6). The package is ready for the IT Security Director to review Section 1; nothing is provisioned until that approval is signed.

1. Verification Method

Every check was run against the four completed Word documents themselves, not against drafts or notes. A script read their tables and the Section 2 JSON Schema directly, and the results were then reviewed by hand. Grounding sources were matched by resource name against the Section 1 hub-and-project table. Handoff fields were matched by name, schema path and base type against the parsed Section 2 schema. Failure nodes were traced by searching each Section 4 architecture point for the agents, tools, escalations and handoff elements named in Sections 2 and 3.

Package section	Document checked	Items read by the check
Section 1: Foundry Workspace Blueprint	Step2_Section1_Foundry_Workspace_Blueprint.docx	Item 2 hub-and-project table; item 3 RBAC; Table 4a managed identities
Section 2: Agent Operational Specifications	Step3_Section2_Agent_Operational_Specifications.docx	Item 1 inventory; item 4 JSON Schema; item 5 escalations; item 8 grounding alignment
Section 3: Multi-Agent Governance and A2A Security	Step4_Section3_MultiAgent_Governance_A2A_Security.docx	Items 3.1 to 3.5 (handoff, schema, redaction, escalation branch)
Section 4: Workflow Stabilization Assessment	Step5_Section4_Workflow_Stabilization_Assessment.docx	Item 2 failure nodes; item 4 observability spans

2. Check 1: Grounding Sources Reside in the Agent's Project

Agent (Section 2)	Grounding source (Section 2, item 8)	Project holding the source (Section 1, item 2)	Agent Project (Section 2)	Result
Claim Intake / Extraction Agent	claims-inbox	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Claim Intake / Extraction Agent	di-apex-claims-prod	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Claim Intake / Extraction Agent	kb-claims-intake-guide	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Claim Intake / Extraction Agent	stapexclaimsintake	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Policy Cross-Reference Agent	kb-claims-policy-wordings	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Policy Cross-Reference Agent	sqldb-apex-policy	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Policy Cross-Reference Agent	vw_policy_coverage	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Fraud and Risk Specialist	apim-apex-integration-prod	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Fraud and Risk Specialist	kb-claims-fraud-indicators	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Fraud and Risk Specialist	sqldb-apex-policy	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Fraud and Risk Specialist	vw_claim_history_36m	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**
Settlement Specialist	kb-claims-settlement-templates	proj-apex-claims-processing	proj-apex-claims-processing	**PASS**

Result: 12 of 12. Every grounding source used by a Claims agent, including the four Foundry IQ knowledge bases on `srch-apex-claims-prod`, is listed only in the `proj-apex-claims-processing` row of Section 1. None appears in the hub row (which has no data connections by design) or in the Legal or Finance rows. Tools such as `calculate_settlement` and `score_fraud_signals` are Project-hosted functions, not data sources.

3. Check 2: A2A Handoff Schema Uses Only Section 2 Field Names

Handoff field (Section 3, item 3.3)	Type in Section 3	Path in Section 2 ClaimCaseOutput v1.0	Type in Section 2	Result
claim_id	string	claim_id	string	**PASS**
correlation_id	string	correlation_id	string	**PASS**
schema_version	string	schema_version	string	**PASS**
fraud_risk_score	number (float 0.00–1.00)	fraud_assessment.fraud_risk_score	number	**PASS**
risk_classification	string (enum LOW | ELEVATED)	fraud_assessment.risk_classification	string	**PASS**
fraud_signals	array<string> (FS-NNN codes)	fraud_assessment.fraud_signals	array	**PASS**
public_records_check_status	string (enum COMPLETED | CACHED)	fraud_assessment.public_records_check_status	string	**PASS**
redacted_claim_summary	string (≤1,500 chars)	redacted_claim_summary	string	**PASS**
coverage_status	string (enum COVERED | PARTIAL)	policy_cross_reference.coverage_status	string	**PASS**
coverage_limit_usd	number	policy_cross_reference.coverage_limit_usd	number	**PASS**
deductible_usd	number	policy_cross_reference.deductible_usd	number	**PASS**
applicable_exclusions	array<string>	policy_cross_reference.applicable_exclusions	array	**PASS**
claimed_amount_usd	number	extracted_claim.claimed_amount_usd	number	**PASS**
loss_type	string (enum)	extracted_claim.loss_type	string	**PASS**
date_of_loss	string (date)	extracted_claim.date_of_loss	string	**PASS**
property_location_state	string (2-letter)	extracted_claim.property_location_state	string	**PASS**
policyholder_token	string (PH-TKN-…)	pii_tokens.policyholder_token	string	**PASS**
policy_number_token	string (POL-TKN-…)	pii_tokens.policy_number_token	string	**PASS**
settlement_hold_required	boolean	routing.settlement_hold_required	boolean	**PASS**

Result: 19 of 19. Every handoff field exists in `ClaimCaseOutput` v1.0 (40 properties) at the path shown, with a matching base type. Where Section 3 narrows an enum (for example `risk_classification`: LOW | ELEVATED), the values are a valid subset of the Section 2 enum, and the narrowing is what enforces the escalation branch. No field was invented for the handoff; A2A envelope fields (`messageId`, `contextId`, `traceparent`) are protocol fields and are kept out of the business payload.

4. Check 3: Failure Nodes Trace to Section 2 Agents and Section 3 Handoffs

Failure node	Section 4 architecture point	Trace to Section 2	Trace to Section 3	Result
FN-001	External API dependency in the fraud and public-records check	Fraud and Risk Specialist; tool public_records_lookup (Part B, pillar 4); ESC-005	Routing gate INDETERMINATE branch to sbq-external-verification-hold (items 1 and 3.5)	**PASS**
FN-002	Settlement Specialist output validation	Settlement Specialist; settlement block (letter_validation_status); ESC-007	Validation gate after Settlement (Figure 1); decomposition row 4 (ESC-007)	**PASS**
FN-003	Agent interaction observability at the A2A handoff Fraud and Risk Specialist → Settlement Specialist, brokered by the Claims Workflow Controller (Section 3, items 3.2 and 3.4), plus the invoke_agent spans of all four agents.	Fraud and Risk Specialist and Settlement Specialist; correlation_id typed input	A2A handoff Fraud and Risk Specialist → Settlement Specialist; items 3.2 (correlation, audit trace) and 3.4	**PASS**

Result: 3 of 3. Each failure node names the Section 2 agent it sits on. The error-handling pattern for each node is the one the checklist requires: circuit breaker for the external API, validation failure branch for Settlement output, observability instrumentation for the handoff. Its fallback lands on a queue or escalation that Sections 2 and 3 already define.

5. Additional Cross-Section Checks

Check	Evidence from the documents	Result
Fraud thresholds identical in Sections 2, 3 and 4	Every comparison against fraud_risk_score in the package uses only: 0.60, 0.85	**PASS**
Handoff payload size stated consistently (19 fields)	Section 3 item 3.3 lists 19 fields; "19-field" stated in Section 3, Section 4	**PASS**
Each budget line has its own model deployment inside its own Project (FinOps meter)	aoai-apex-claims-prod, aoai-apex-legal-prod and aoai-apex-finance-prod each appear in their own Project row only	**PASS**
Large-loss threshold ($250,000) consistent	Found in Section 2, Section 3, Section 4; no other claim-value threshold used	**PASS**
Correlation ID format identical	{claim_id}.{workflow_run_id} found in Section 2, Section 3	**PASS**
Managed identities used in Sections 2 to 4 are defined in Section 1 (Table 4a)	Used: uami-claims-fraud-risk, uami-claims-workflow-controller	**PASS**
Span names cited in Sections 1 to 3 exist in the Section 4 observability specification	Cited: a2a.handoff, fraud.routing_decision, invoke_agent, pii.redaction, settlement_output.validation	**PASS**
Escalation queues in Sections 3 and 4 are defined in Section 2	Used: sbq-adjuster-review, sbq-external-verification-hold, sbq-large-loss-approval, sbq-siu-fraud-review	**PASS**
Escalation IDs in Sections 3 and 4 are defined in Section 2 (item 5)	Used: ESC-001, ESC-002, ESC-003, ESC-004, ESC-005, ESC-006, ESC-007	**PASS**
Legal Project shares no data source with the Claims Project	Legal sources: aoai-apex-legal-prod, kb-legal-regulatory-library, srch-apex-legal-prod	**PASS**
Finance Project shares no data source with Claims or Legal	Finance sources: aoai-apex-finance-prod, fda-claims-kpi, ws-apex-claims-analytics	**PASS**
No Legal resource is used as grounding in Section 2	Legal identifiers in Section 2: srch-apex-legal-prod	**PASS**
Legal RBAC scope excludes the Claims Project	Legal scope: proj-apex-compliance-legal only	**PASS**
All four task agents sit in the Section 1 Claims Project	Project Boundary column: proj-apex-claims-processing	**PASS**

6. Issues Found and Resolved

#	Finding	Sections	Resolution	Status
1	Section 2 typed inputs (claim_id, policyholder_id, submission_channel) are read from blob metadata by the Claims Workflow Controller, but Section 1 Table 4a gave uami-claims-workflow-controller no access to claims-inbox. Without a fix the trigger step would fail authorization.	1, 2	Added **Storage Blob Data Reader on claims-inbox** to the controller identity in Section 1, Table 4a.	**Fixed**
2	ESC-004 (no coverage) could have routed to coverage counsel in Legal, which would give Legal access to claims data and breach Section 1.	1, 2	ESC-004 routes to the **Claims Coverage Review team in Operations** (sbq-coverage-review). Section 2 states that Legal is never a destination.	Resolved by design
3	Audit consumers for Claims telemetry could have included Legal, for the same reason.	1, 4	Section 4 gives read access to **Claims Compliance and Internal Audit** only. Legal receives curated audit reports.	Resolved by design
4	The Settlement Specialist needs the insured's name and address to write a letter, but the Section 3 redaction rules remove them from the handoff.	1, 3, 4	Merge tokens plus a non-LLM **Letter Rendering Service** with its own identity (uami-claims-letter-render, Table 4a) re-insert PII after validation (Section 4, rule V3).	Resolved by design
5	The Section 4 cached fallback for the public-records API could weaken scrutiny of large losses.	2, 3, 4	CACHED results are **not accepted when claimed_amount_usd > $250,000**. This is stated in Section 2 (ESC-001), Section 3 (standard path) and Section 4 (FN-001).	Resolved by design

7. Sign-off Readiness

Package section	Status	Next action
Section 1: Foundry Workspace Blueprint	Complete; consistency checks pass	IT Security Director reviews and signs the approval block in Section 1, item 6. **Provisioning stays blocked until then.**
Section 2: Agent Operational Specifications	Complete; consistency checks pass	SIU confirms fraud-score calibration before go-live (Section 4, risk R4).
Section 3: Multi-Agent Governance and	Complete; consistency checks	Identity team registers app roles Handoff.Submit and

Package section	Status	Next action
A2A Security	pass	Settlement.Calculate at provisioning.
Section 4: Workflow Stabilization Assessment	Complete; consistency checks pass	Tune circuit-breaker thresholds against a vendor latency baseline during the pilot (risk R3).

Re-run rule: any change to a threshold, field name, Project boundary, identity or span name in one section triggers a re-run of all checks in this record before resubmission, because those values are shared across all four sections.

Submission Checklist

☑ Every grounding data source in Section 2 resides in the same Project as the agent that references it (Section 1).

☑ The A2A handoff schema in Section 3 uses only field names defined in the task agent output schema in Section 2.

☑ The failure nodes in Section 4 trace to specific agents and handoffs described in Sections 2 and 3.